

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost &gt; 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)
- ◆ SAC-Lagrangian (penalty-based)
- ▲ SAC + fixed penalty (penalty-based)
- ▼ SDAC (multi-constraint SOTA)
- ⊕ SRCPO (multi-constraint SOTA)
- typed Pareto frontier (soft  $\leq 25$ )
- ⋯ SAC-Pen coefficient sweep
- - - FDPI  $p_f$  sweep
- - - SAC-Lag cost-limit sweep